

Dixit Sabharwal

Software Engineer

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Profile

I am a full-stack developer who builds functional interfaces and maintainable APIs that adhere to open standards. I deploy these solutions across various environments; from the AWS cloud, to on-prem Kubernetes clusters, and bare-metal VMs.

I'm dedicated to learning and building systems that are reliable and practical. I am particularly interested in working on projects or with companies that support open-source software or work in the public sector. I'm convinced of the importance of making quality software accessible to everyone.

Experience

Software Engineer @ AWS Billing (Amazon Web Services)

Nov 2024 - Present
Berlin, Germany

- Develop and operate the system of record for the AWS product catalog — 90M+ requests per day across 6 regions (US, EU, China) at 99.95%+ availability — including 24/7 on-call, incident response and post mortems.
- Cut the internal DataWarehouse data-freshness SLA from 8 hours to under 10 minutes by replacing snapshot exports with a CDC pipeline writing to an Apache Iceberg catalog.
- Migrated 90% of the service's data from a MySQL cluster to DynamoDB with zero downtime, holding the 99.95% availability SLA throughout.
- Automated zero-touch creation of new production environments, enabling the service to be stood up in a new AWS region with 3 weeks of passive work from a single engineer.
- Contributed to a web console used by the product owners of 250+ Amazon services to configure and price their catalog and offerings, with built-in A/B testing and approval workflows.

Software Developer (Internship) @ Faculté IC, EPFL

June 2023 - Jul 2023
Lausanne, Switzerland

- Part of team that set-up an automatic grading system for programming assignments. Through a Moodle plugin and Kubernetes cluster the system allowed for easy scalability and fault-tolerance during peak times of the semester.
- Created a controller (using Spring and Java) that runs each submission as a Job on the Kubernetes cluster. The controller sets up the env for the grading task, and returns the outcomes to Moodle.
- Set-up CI/CD pipeline using GitLab CI that automatically deploys the controller and Moodle in a staging environment, for quick integration testing and feedback.

Data Analyst (Internship) @ Taskbase

Jul 2023 - Sep 2023
Zurich, Switzerland

- Built CLI tool which extracts data from CSV and Excel sources provided by clients and passes it through the Taskbase API, to speed up the onboarding process.
- Generated sample data for QA tasks, through prompt engineering and OpenAI GPT models, to compensate for the lack of real data from new clients.

Software Development Intern @ AWS/Amazon

Aug 2022 - Feb 2023
Berlin, Germany

- Built a management console to empower ongoing migration of service to a new data model and store.
- Console is a web-based UI to view and search through data, also compare between old and new versions, and trigger different tasks in the migration pipeline.
- Used React and TypeScript for the website UI.
Used the AWS serverless stack (Lambda, API Gateway, DynamoDB) for the backend.
Managed infrastructure in AWS CDK.

Teaching Assistant @ Faculty of EEMCS, TU Delft

Sep 2019 - Jun 2021
Delft, Netherlands

- Teaching Assistant for > 10 different courses for the Computer Science bachelors program.

Web Developer @ Dizconto Ltd.

Apr 2020 - Jul 2020
Den Haag, Netherlands

- Built a management console frontend for Dizconto, which provided social media marketing integrations.
- Planned website development, converted mockups into usable web presence with TypeScript, React and NodeJS.
- Also worked on implementing a development and deployment (CI/CD) pipeline using Terraform and AWS.

Education

M.Sc. in Computer Science

EPFL
Sep 2021 - July 2024

- Coursework in software security and privacy, compilers and type-systems, and large-scale data handling.
- Thesis on "Analysis of AI augmented writing tools for contemporary literature".

B.Sc. in Computer Science Engineering

TU Delft
Sep 2018 - July 2021

- Graduated cum laude with a GPA of 8.8.
- Thesis on "Formal verification of functional programs in Haskell and Agda".
- Minor in Quantum Science and Information.

Skills

Languages: Java/Kotlin, Rust,
Javascript/Typescript, Python

UI: React.js, TanStack

Server Frameworks: Spring Boot, Hono,
Axum/Tower

Data Analysis: Pandas, Scikit-learn

Database Technologies: SQLite, PostgreSQL,
MySQL, DynamoDB

Data Engineering: Apache Spark, Apache
Iceberg, DuckDB

Deployment Environments: AWS, Azure,
Kubernetes (k8s), Docker

AWS Serverless: Lambda, ECS, Step Functions,
Glue

CI/CD: GitLab CI